

JEE MAIN REVISION MASTER: WAVE OPTICS SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 178 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Wave Optics - Core Application	178 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Wave Optics - Core Application (Very High Yield — 178 PYQs)

Core Principle: Key problem pattern in Wave Optics.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "The maximum intensity in a Young's double slit experiment is I_0 . Distance between the slits (d) is 5λ , where λ is $t...$ " [Ans: B]
- [2026 Morning Shift] Q#1 (MCQ): "A telescope with objective diameter R is used to observe a distant star emitting light of wavelength 500 nm , at a resolution of $5\times...$ " [Ans: B]
- [2026 Evening Shift] Q#1 (MCQ): "When an unpolarized light falls at a particular angle on a glass plate (placed in air), it is observed that the reflected beam is linearly p..." [Ans: D]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 5th April Evening Shift (Q#1)

The maximum intensity in a Young's double slit experiment is I_0 . Distance between the slits (d) is 5λ , where λ is the wavelength of light used. The intensity of the fringe, exactly opposite to one of the slits on the screen, placed at $D=10d$ is _____.

- (A) $\frac{I_0}{4}$
- (B) $\frac{I_0}{2}$ **[Correct]**
- (C) I_0
- (D) $\frac{3I_0}{4}$

Correct Answer: B

Solution Analysis: A point exactly opposite to one of the slits is at a distance $y=(d/2)$ from the central maximum. The slit separation is given as $d=5\lambda$. The distance between slits and screen is $D=10d=50\lambda$. And the vertical distance of required point from the central point is $y=(d/2)=2.5\lambda$. For a point on the screen, path difference is approximately: $\Delta x=y \cdot (d/D)$. Substituting the given values: $\Delta x=((d/2)) \cdot (d/10d)=(d/20)$. Substituting $d=5\lambda$: $\Delta x=(5\lambda/20)=(\lambda/4)$. The relation between phase difference